UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 08/06/2017*

EXERCISE

Given a random variable Z with probability density function (pdf) uniform between 0 and 2α, , i.e. $f_Z(z) = \frac{1}{2\alpha} rect\left(\frac{z-\alpha}{2\alpha}\right)$, where $\alpha > 0$.

Assuming that we observe N independent realizations of Z:

(1) plot the pdf of Z and the likelihood function (LF) $L(\alpha)$;

(2) derive the maximum likelihood (ML) estimator of the deterministic parameter α and its probability density function;

(3) derive the bias and the mean square error (MSE) of the ML estimator of α and establish if the estimator is consistent;

(4) derive the method of moments (MM) estimator of parameter α, its bias and MSE, and establish if the estimator is consistent;

(5) plot and compare the MSE's of the two estimators; comment on the pros and cons of the two estimators;

(6) find what is the minimum value of the data size N that guarantees that the standard deviation of the relative error is lower than 10%.

Note 1: The cumulative distribution function (CDF) of Z is given by:

$$F_Z(z;\alpha) = \int_{-\infty}^{z} f_Z(x;\alpha)dx = \begin{cases} 0, & z \le 0 \\ \frac{z}{2\alpha}, & 0 < z \le 2\alpha \\ 1, & z > 2\alpha \end{cases}$$

Note 2: If a random variable (r.v.) Y is uniformly distributed between a and b, it holds true that:

$$E\{Y\} = \frac{b+a}{2}, \quad \text{var}\{Y\} = \frac{(b-a)^2}{12}.$$

Solution D.1 - D.2

$$L\left(\alpha|z_1\right) = f_Z(z_1;\alpha) = \begin{cases} 1/(2\alpha), & 0 \le z_1 \le 2\alpha \\ 0, & otherwise \end{cases}$$

$$L\left(\alpha|\mathbf{z}\right) = \prod_{n=1}^{N} f_Z(z_n;\alpha) = \begin{cases} 1/(2\alpha)^N, & 0 \le z_n \le 2\alpha \quad \forall n \\ 0, & otherwise \end{cases}$$

$$L\left(\alpha|\mathbf{z}\right) = \begin{cases} 1/(2\alpha)^N, & 0 \le \max\{z_n\} \le 2\alpha \\ 0, & otherwise \end{cases}$$

$$L\left(\alpha|\mathbf{z}\right) = \frac{1}{(2\alpha)^N} u\left(\alpha - \frac{\max\{z_n\}}{2}\right) \quad \rightarrow \quad \hat{\alpha}_{ML} = \frac{1}{2}\max\{z_n\}$$

$$X \triangleq \hat{\alpha}_{ML} = \frac{1}{2}\max\{z_n\}$$

$$\begin{aligned} F_X(x) &= \Pr\{X \le x\} = \Pr\left\{\frac{1}{2}\max\{z_n\} \le x\right\} = \Pr\{\max\{z_n\} \le 2x\} \\ &= \Pr\{z_1 \le 2x, \cdots, z_N \le 2x\} = \prod_{n=1}^{N} \Pr\{z_n \le 2x\} = \left(F_Z(2x)\right)^N \end{aligned}$$

$$f_X(x) = \frac{dF_X(x)}{dx} = 2N\left(F_Z(2x)\right)^{N-1} f_Z(2x)$$

The pdf of the ML estimate is given by::

$$f_{\hat{\alpha}_{ML}}(x) = f_X(x) = 2N\left(\frac{2x}{2\alpha}\right)^{N-1} \cdot \frac{1}{2\alpha} rect\left(\frac{2x-\alpha}{2\alpha}\right) = \frac{N}{\alpha^N} x^{N-1} rect\left(\frac{x-\alpha/2}{\alpha}\right)$$

Solution D.3

$$E\{\hat{\alpha}_{ML}\} = \int_{-\infty}^{+\infty} x f_X(x) dx = \frac{N}{\alpha^N}\int_0^{\alpha} x^N dx = \frac{N}{(N+1)}\alpha$$

$$E\left\{\left(\hat{\alpha}_{ML}\right)^2\right\} = \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \frac{N}{\alpha^N}\int_0^{\alpha} x^{N+1} dx = \frac{N}{(N+2)}\alpha^2$$

Bias: $b\{\hat{\alpha}_{ML}\} = E\{\alpha - \hat{\alpha}_{ML}\} = \alpha - \frac{N}{(N+1)}\alpha = -\frac{1}{(N+1)}\alpha$

$$MSE\{\hat{\alpha}_{ML}\} = \text{var}\{\hat{\alpha}_{ML}\} + \left(b\{\hat{\alpha}_{ML}\}\right)^2 = E\left\{\left(\hat{\alpha}_{ML}\right)^2\right\} - \left(E\{\hat{\alpha}_{ML}\}\right)^2 + \left(b\{\hat{\alpha}_{ML}\}\right)^2$$
$$= \frac{N}{(N+2)}\alpha^2 - \left(\frac{N}{(N+1)}\alpha\right)^2 + \left(-\frac{1}{(N+1)}\alpha\right)^2 = \frac{2\alpha^2}{(N+2)(N+1)}$$

The ML estimator is biased, but asymptotically unbiased, and consistent. Note that the calculation of the ML estimate requires a nonlinear search for the maximum of the data.

Solution D.4

$$E\{Z\} = \alpha \quad \rightarrow \quad \hat{\alpha}_{MM} = \frac{1}{N}\sum_{n=1}^{N} Z_n$$

$$E\{\hat{\alpha}_{MM}\} = E\left\{\frac{1}{N}\sum_{n=1}^{N} Z_n\right\} = \alpha$$

$$MSE\{\hat{\alpha}_{MM}\} = E\left\{\left(\hat{\alpha}_{MM} - \alpha\right)^2\right\} = \frac{\text{var}\{Z_n\}}{N} = \frac{(2\alpha)^2}{12N} = \frac{\alpha^2}{3N}$$

The MM estimator is linear (i.e. it is a linear function of the data), it is unbiased and consistent.

Since $MSE\{\hat{\alpha}_{ML}\} = \frac{2\alpha^2}{(N+2)(N+1)} \leq \frac{\alpha^2}{3N} = MSE\{\hat{\alpha}_{MM}\}$, the two MSE's are equal only for N=1, otherwise the ML estimator is more efficient than the MM estimator.

Solution D.6

The minimum value of N that guarantees that the standard deviation of the relative error is lower than 10% is immediately derived:

MM: $\sigma_{\varepsilon_r} = \sqrt{E\{\varepsilon_r^2\}} = \sqrt{E\left\{\left(\frac{\alpha - \hat{\alpha}_{MM}}{\alpha}\right)^2\right\}} = \sqrt{\frac{MSE\{\hat{\alpha}_{MM}\}}{\alpha^2}} = \sqrt{\frac{1}{3N}} \leq \frac{10}{100} \quad \rightarrow \quad N = 34$

ML: $\sigma_{\varepsilon_r} = \sqrt{E\{\varepsilon_r^2\}} = \sqrt{E\left\{\left(\frac{\alpha - \hat{\alpha}_{ML}}{\alpha}\right)^2\right\}} = \sqrt{\frac{MSE\{\hat{\alpha}_{ML}\}}{\alpha^2}} = \sqrt{\frac{2}{(N+2)(N+1)}} \leq \frac{10}{100} \quad \rightarrow \quad N = 13$

Solution D.5 – Plot of the MSE's